



FC with RG – Neuromorphic Computing

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Semester Goals:

- Gain a deeper understanding of SNNs and neuromorphic computing as a whole
- Master snnTorch fundamentals
- Reproduce and extend the results from the thesis “Bridging Neuromorphic Platforms for Customized Recurrent Spiking Neural Networks: Human Activity Recognition from snnTorch to Intel Loihi 2” by Dr. Gianvito Urgese and Dr. Vittorio Fra

Project Overview and Methods:

- Completed all 7 snnTorch tutorials, along with additional tutorials (e.g. exoplanet)
- Hosted weekly “Book Clubs”: Reading weekly SNN-related research papers and discussing them during subteam meetings
- Used Neuromorphic thesis as blueprint to reproduce the results of the paper
 - Preprocessing: preprocessed WISDM data to create training samples
 - Spike Encoding: converting continuous data into SNN spike trains
 - Inhibitory Model: Replicating the SNN's structure and processing the spike train pipeline



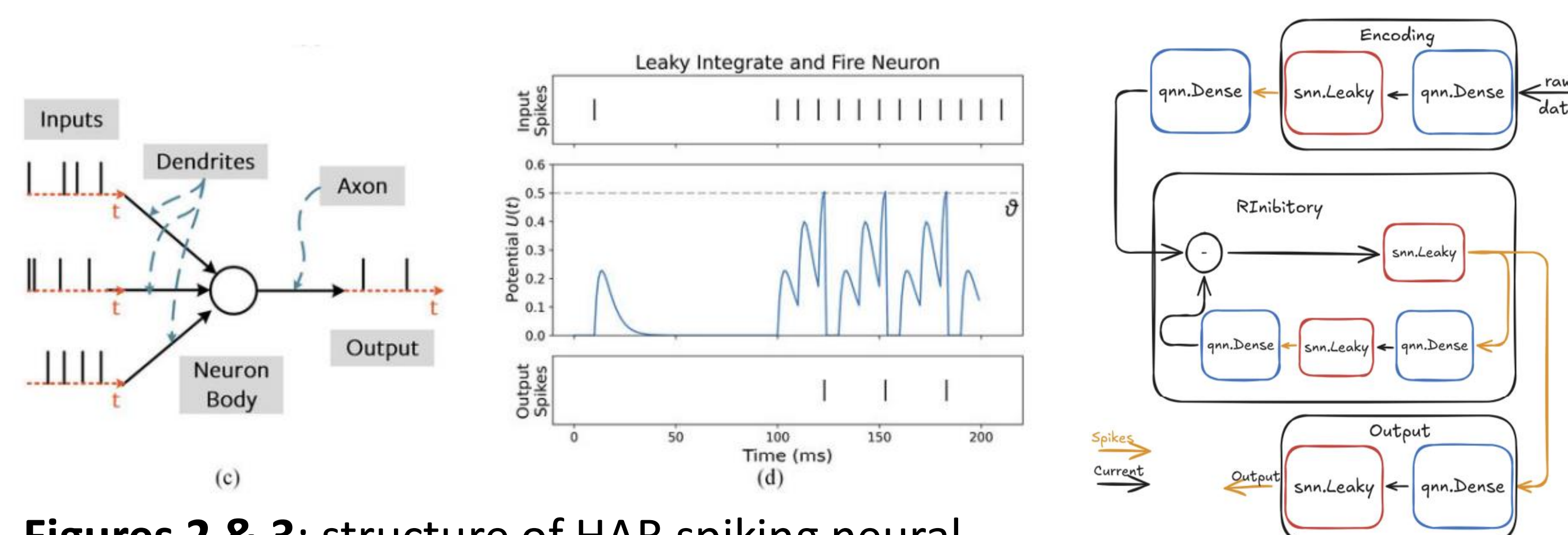
Experiments and Results

Experimentation on trial datasets

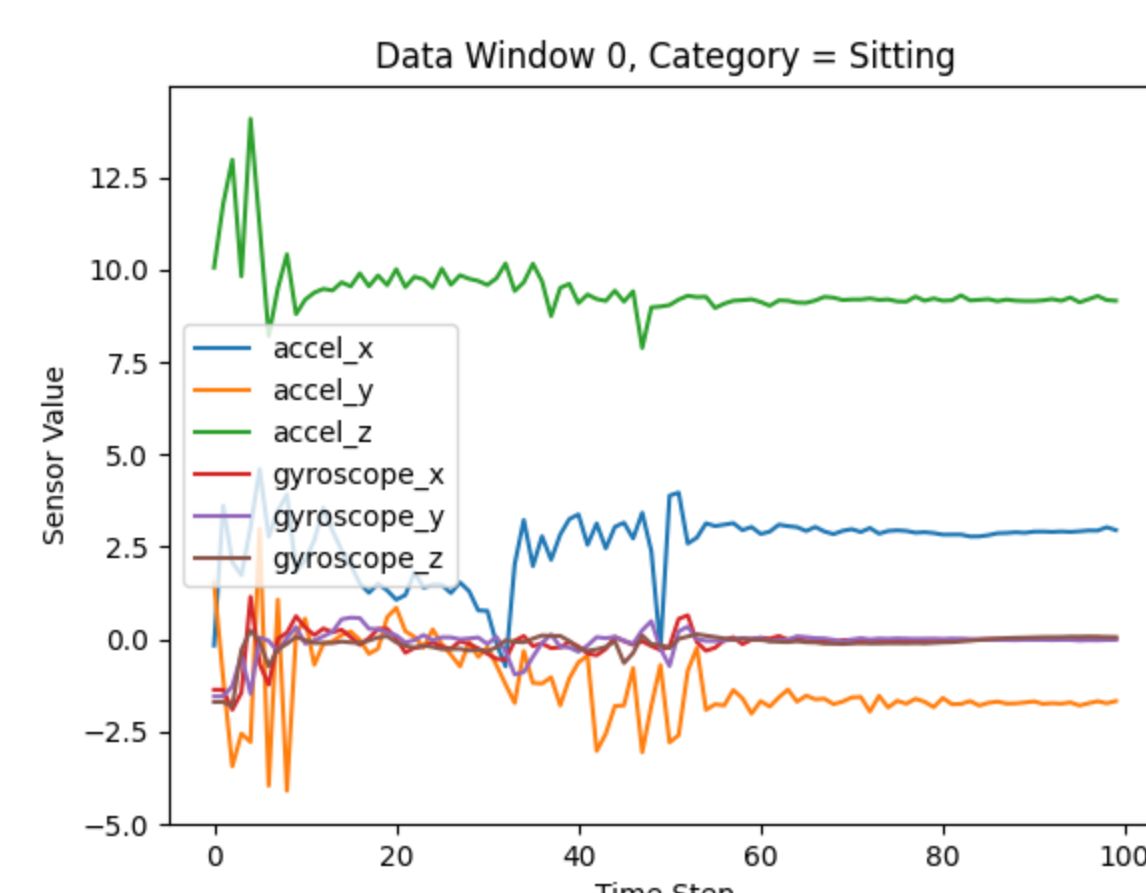


Figure 1: training of exoplanet detection model utilizing light flux of stars

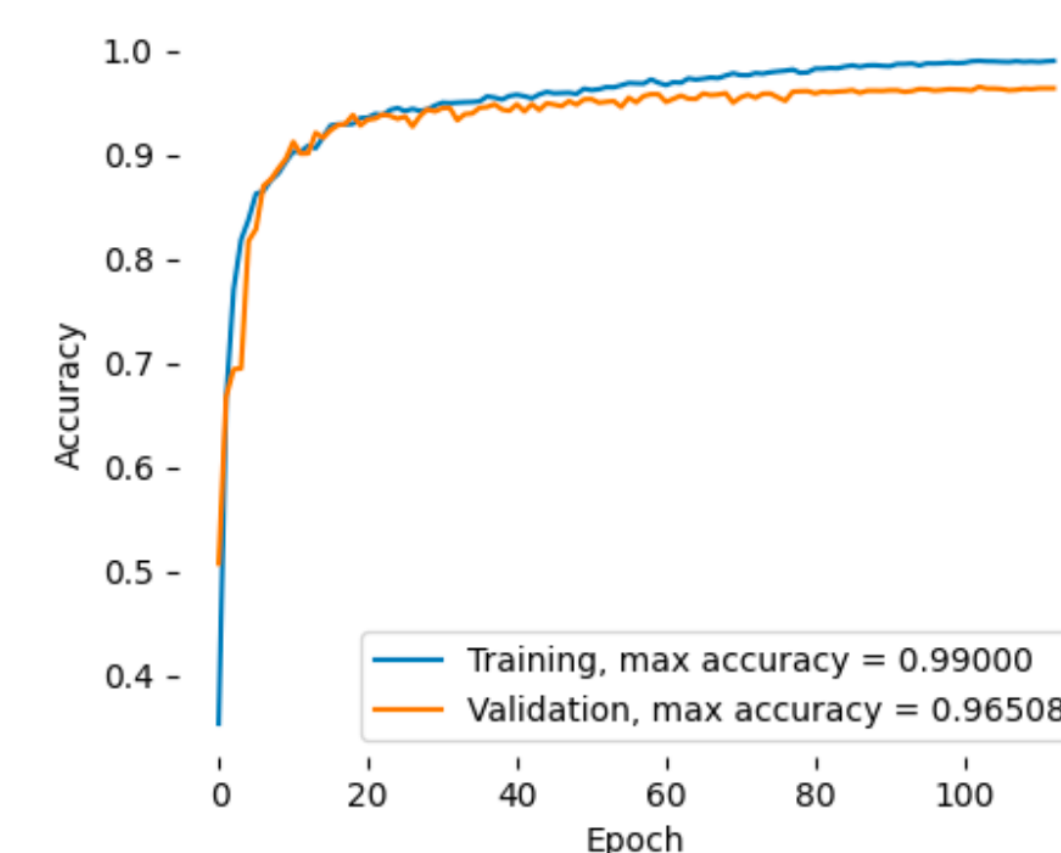
Replication of Human Activity Recognition (HAR) paper



Figures 2 & 3: structure of HAR spiking neural network, sourced from Urgese and Fra's paper



Figures 4 & 5: Example preprocessed training window and model accuracy result for Tesi paper replication



Index	Activity
0	walking
1	jogging
2	stairs
3	sitting
4	standing
5	typing
6	brushing teeth
7	eating soup
8	eating chips
9	eating pasta
10	drinking
11	eating sandwich
12	kicking soccer
13	catch tennis
14	dribbling basketball
15	writing
16	clapping
17	folding clothes

Table 2.1: Complete List of Activities

Figure 6: Categorization of human activity from Tesi paper

Lessons Learned

We learned the fundamentals of spiking neural networks ("SNNs"): spike encoding techniques, neuron models, as well as network creation and training. Additionally, we covered applications of SNNs, their benefits and weaknesses, and their implementation in hardware.

Next Steps and Future Challenges

- Continue replicating Tesi category classification paper, finish spike encoding and inhibitory layers
- Test on real-world noisy data: WISDM is clean; validate robustness on data with motion artifacts and sensor drift from other devices
- Cross-subject generalization: current model trains/tests on same subjects; evaluate performance on unseen individuals
- Create an SNN to denoise data collected from Nab particle physics experiments
- Apply SNNs to graph traversal algorithms